EXPERIMENT

Construction of the First P7 Mutant Panel

BRANCH	DATE	EXPERIMENTER
PUF	2026-07-28	Xiaoyu Que

Experimental Summary

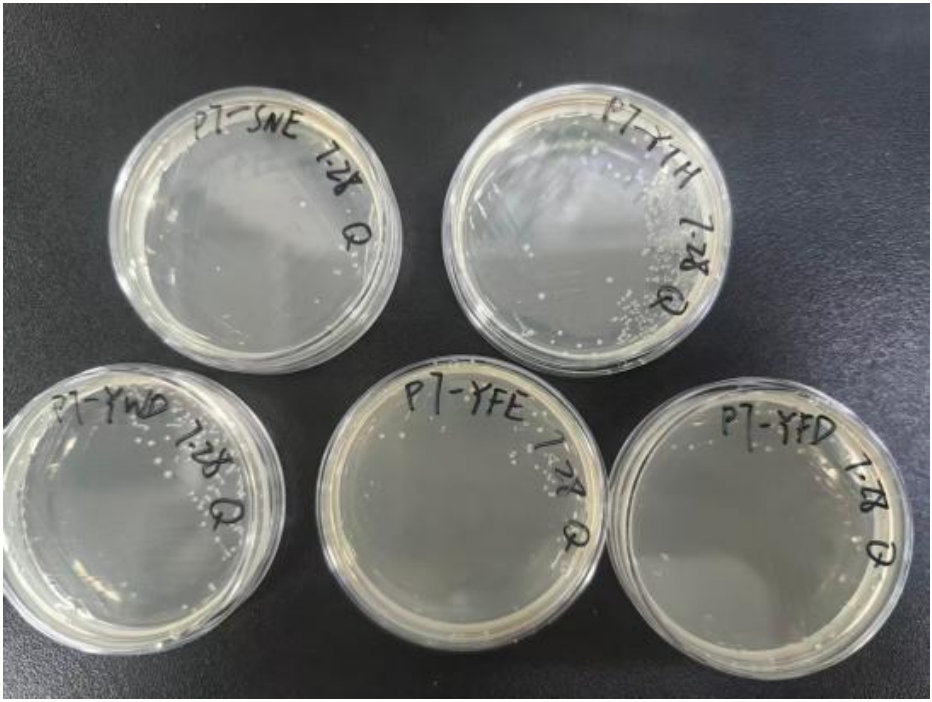
Sixteen P7 variants were amplified, treated with DpnI, gel-purified, and assembled with CE mix V3 at 50 °C for 15 min. The assembly products were transformed into DH5α and plated.

Results

Subsequent sequencing of the initial batch supported THE, SNE, and YFD; the remaining groups showed a 324-bp deletion. Later troubleshooting attributed the deletions to recombination between homologous repeats and reported improvement after shortening assembly from 15 to 5 min.

EXPERIMENT SOURCE IMAGE

Construction of the First P7 Mutant Panel



Selected plates from the July 28 P7 batch: SNE, YTH, YWD, YFE, and YFD. These plates show colony growth from the cloning batch; the sequence-screening outcomes are described in the text.

RELATED PROTOCOLS

M01: PCR; M02: Agarose Gel Electrophoresis and Gel Extraction; M03: Restriction Enzyme Digestion; M04: Homologous Recombination; M05: Transformation & Plating.